

SONAL SUHAS RAO

Data Analyst | Business Intelligence Analyst | Python | SQL | Power BI | Tableau
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WORK EXPERIENCE

Associate Developer

Mitsubishi Pension and Market Services (formerly Link Group), Mumbai

Jul 2022 – Jul 2024

- Developed and maintained .NET MVC middleware applications supporting large-scale Australian investor and pension management systems.
- Optimized SQL Server queries and stored procedures to improve reporting performance and reduce data retrieval times.
- Analyzed business and system requirements in collaboration with cross-functional stakeholders, translating operational needs into technical solutions. Improved data quality and operational efficiency by identifying and resolving system performance bottlenecks.
- Supported production systems handling high-volume financial and investor records while ensuring reliability and compliance requirements.

Software Intern

Grassdew IT Solutions Pvt. Ltd.

Sep 2021 – Jan 2022

- Developed data science and cybersecurity case-study projects using real-world datasets and business scenarios.
- Performed exploratory data analysis and applied machine learning techniques to solve practical business problems.
- Created technical documentation and project reports to communicate findings and support knowledge sharing.

Data Science Intern

LetsGrowMore

Aug 2021 – Sep 2021

- Built machine learning models using Python and Scikit-learn on real-world datasets.
- Performed data cleaning, feature engineering, and model evaluation to improve predictive performance.
- Created visualizations and dashboards to communicate analytical findings and business insights.
- Presented project outcomes and recommendations to mentors and stakeholders.

PROJECT EXPERIENCE

VehicleIQ – AI-Powered Vehicle Intelligence Platform | Python, PyTorch, OpenCV, FastAPI

- Developed a computer vision system capable of identifying vehicle make, model, colour, body type, and damage from images.
- Built multiple deep learning models achieving strong performance across vehicle attribute classification tasks.
- Implemented an end-to-end pipeline for automated vehicle analysis where registration plates were unavailable.

Sales Insight Dashboard | Power BI, Tableau

- Designed interactive dashboards to analyze sales performance and business metrics
- Built dynamic visualizations with drill-down capabilities using real-time data sources
- Identified revenue-driving products and market trends, supporting data-driven business decisions

Customer Churn Prediction | Python, Scikit-learn, Pandas

- Developed machine learning models to predict customer churn
- Applied preprocessing, feature engineering, and SMOTE with Random Forest and Decision Tree models
- Identified key churn drivers and enabled targeted retention strategies to reduce revenue loss

Airbnb Data Pipeline | Airflow, dbt, PostgreSQL, Google Cloud

- Designed a scalable ELT pipeline for Airbnb and Census datasets
- Implemented Medallion architecture using Airflow for orchestration and dbt for transformation and testing
- Delivered clean, analytics-ready data models and generated insights on revenue trends and host behaviour

NYC Taxi Fare Prediction & Big Data Analysis | PySpark, SQL, Scikit-learn

- Built a big data pipeline and predictive model for taxi fare estimation
- Processed 10+ years of data using PySpark and developed regression models with feature engineering
- Improved prediction accuracy (~81% RMSE reduction) and uncovered insights on demand, pricing, and driver efficiency.

EDUCATION

University of Technology Sydney, Ultimo, NSW | Master of Data Science and Innovation | July 2024 - May 2026

University of Mumbai, India | B.E. in Electronics and Telecommuniucations | GPA: **8.72/10** | August 2018 - May 2022

SKILLS

- **Programming:** Python, SQL
- **Data Analysis:** Pandas, NumPy, Data Cleaning, EDA, Feature Engineering
- **Machine Learning:** Regression, Classification, Clustering, Random Forest, Decision Trees, Model Evaluation
- **Deep Learning & AI:** PyTorch, Neural Networks, Computer Vision, NLP, RAG, LangChain
- **Data Visualisation:** Tableau, Power BI
- **Data Engineering:** Airflow, dbt, PostgreSQL, Google Cloud Platform
- **Tools:** Git, GitHub, Jupyter Notebook, VS Code, Docker, FastAPI

CERTIFICATIONS

- **Machine Learning & Image Processing** – NASSCOM FutureSkills
- **Python Programming** – IIT Bombay
- **Data Science Nanodegree** – Preplnsta